AI-Based Interventions and Perceived Bias in News Coverage of China: Primary Analysis

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Abstract

We study the effects of AI-based interventions on perceptions of bias in news coverage of China using a pre-registered online experiment ($N = 1{,}017$; US, UK, and Canada). Participants read a news article and were randomly assigned to one of three treatments varying in interactivity: an interactive AI chatbot, a tailored AI-generated comment, or a generic static message. Taking the static message as the baseline, the chatbot significantly increases perceived bias against China by approximately 8.5 points on a 0–100 scale ($p < 0.001$). The comment produces an intermediate effect (+5.3 points, $p = 0.008$). No significant treatment differences are found for perceived trustworthiness or sharing intentions; both the chatbot and comment increase perceived one-sidedness. Treatment effects are consistent across all three countries. Exploratory analysis suggests that the verification motive—participants' desire to verify the article further—is the information-processing channel most strongly associated with bias perception change, particularly in the interactive treatments.

1 Experimental Design

1.1 Overview

The experiment studies how AI-generated interventions of varying interactivity affect participants' perceptions of a news article about China. All participants read the same article ("China's Long-Promised Consumer Boom...") and were randomly assigned to one of three treatment arms:

- **Static** ($n = 294$, 28.9%): A generic static message. This is the baseline condition.

- **Comment** ($n = 331$, 32.5%): A single, tailored AI-generated comment personalized to the participant's stated opinion.
- **Chatbot** ($n = 392$, 38.5%): Interactive AI chatbot conversation with up to 3 rounds of dialogue.

1.2 Outcomes

Four outcome measures were elicited both before and after treatment on a 0–100 continuous scale:

1. **Primary outcome**: Perceived bias against China.
2. **Secondary outcomes**: Perceived trustworthiness, perceived one-sidedness, and sharing likelihood.

The estimand of interest is the change score (post − pre) for each measure.

1.3 Estimands

The *primary estimand* is the intent-to-treat (ITT) effect of the Chatbot versus the Static condition on the change in perceived bias against China, controlling for pre-treatment bias. Secondary estimands include the Comment versus Static contrast and all contrasts on the three secondary outcomes.

1.4 Sample

Table 1 reports the participant flow. The sample comprises participants from three English-speaking countries recruited via Prolific ($N = 1{,}017$). No observations are excluded. All participants passed the attention check.

Table 1: Participant Flow

	Static	Comment	Chatbot	Total
US	87	106	140	333
UK	97	109	131	337
Canada	110	116	121	347
Total	294	331	392	1,017

Notes: No exclusions applied. All participants passed the recall-based attention check.

2 Summary Statistics and Balance

Table 2 reports summary statistics by treatment group. Panel A presents demographic characteristics with p-values from OLS regressions of each covariate on treatment indicators. Most covariates are well-balanced. Trust in AI ($p = 0.001$) and income ($p = 0.041$) show some imbalance; both are controlled for in the full specification with no meaningful change in estimates. Panel B reports the race distribution. Pre-treatment outcome levels are well-balanced across arms (all $p > 0.50$; not shown).

Table 2: Summary Statistics

	Static (1)	Comment (2)	Chatbot (3)	Total (4)	p-value (5)
Panel A: Demographics					
Age	43.77	41.87	42.98	42.85	0.329
	(16.29)	(15.86)	(15.45)	(15.83)	
Female (%)	0.50	0.55	0.47	0.51	0.110
Income (1–10)	5.21	5.48	5.54	5.42	0.041
	(1.79)	(1.75)	(1.60)	(1.71)	
Education	3.10	3.00	3.07	3.06	0.513
	(1.30)	(1.00)	(1.04)	(1.11)	
Pol. orientation (1–7)	3.72	3.73	3.96	3.82	0.058
	(1.52)	(1.54)	(1.56)	(1.54)	
Trust in AI (1–7)	4.12	4.23	4.48	4.29	0.001
	(1.38)	(1.49)	(1.33)	(1.41)	
Favorable to China (1–5)	3.12	3.11	3.19	3.14	0.424
	(0.84)	(0.94)	(0.85)	(0.87)	
Panel B: Race/Ethnicity (%)					
White	68.0	70.1	67.9	68.6	
Black	9.2	8.2	13.0	10.3	
Hispanic	4.1	3.3	3.1	3.4	
Asian	12.9	11.8	10.7	11.7	
Mixed/Other	5.8	6.6	5.4	5.9	
Observations	294	331	392	1017	

Notes: Standard deviations in parentheses. Column (5) reports p-values from OLS regressions of each covariate on treatment indicators with robust standard errors.

3 Treatment Effects

3.1 Primary Result: Perceived Bias Against China

Figure 1 displays mean within-arm change scores (post − pre) with 95% confidence intervals. The bottom-left panel shows the key pattern: the Chatbot arm exhibits a large increase in perceived bias against China, the Comment arm an intermediate increase, and the Static arm essentially no change. All three arms show comparable declines in perceived trustworthiness (top-left).

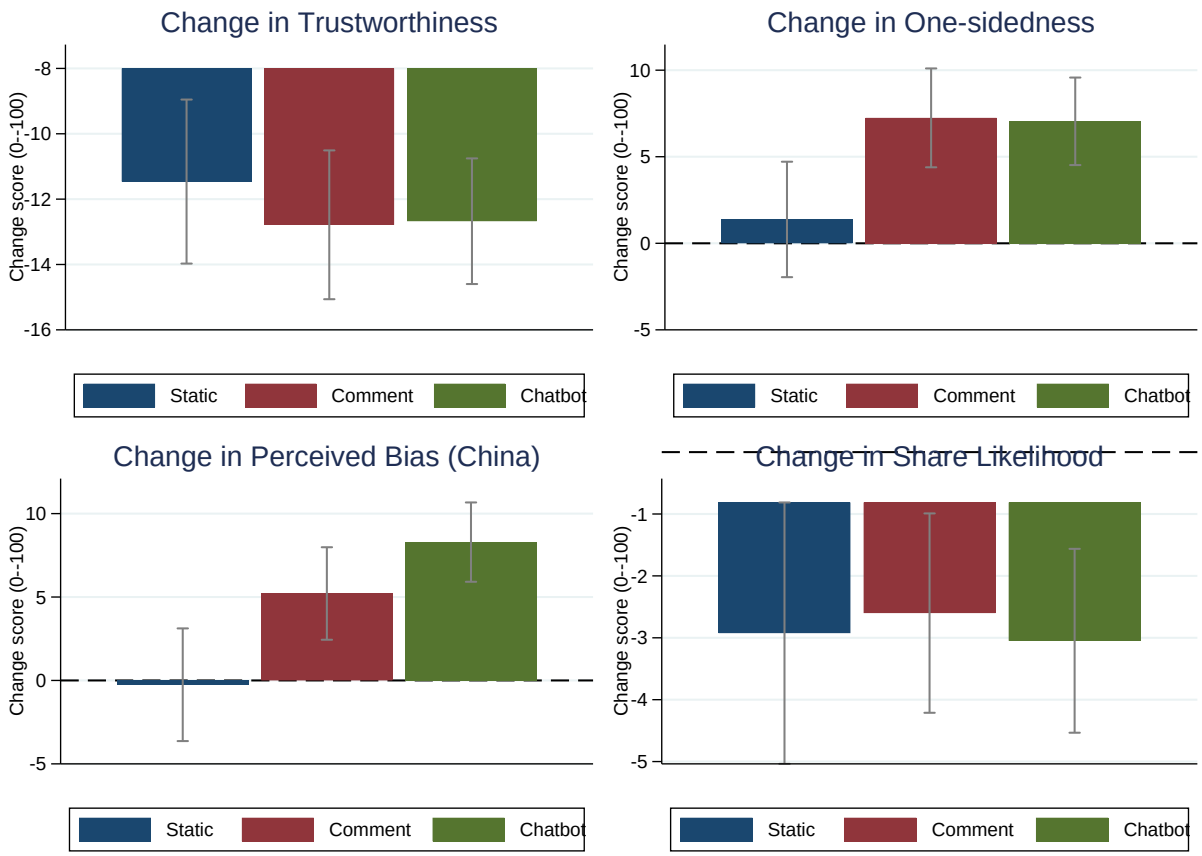

Figure 1: Changes in News Perceptions by Treatment (with 95% CIs)

Table 3 presents the primary estimand—the between-arm treatment effect relative to Static—across three specifications. The Chatbot effect is large and robust across all specifications ($p < 0.001$). With the addition of Canada to the sample, the Comment effect is now also statistically significant in the preferred specification (column 2: $\beta = 5.27$, $p = 0.008$).

Table 3: Treatment Effects on Perceived Bias Against China

	Change in Perceived Bias (0–100)		
	(1)	(2)	(3)
Chatbot	8.549***	8.515***	8.483***
	(2.104)	(1.905)	(1.919)
Comment	5.470**	5.290***	5.000**
	(2.226)	(1.996)	(2.007)
Pre-treatment bias		-0.423***	-0.427***
		(0.031)	(0.032)
Age			-0.104**
			(0.050)
Female			1.744
			(1.510)
Education			0.354
			(0.710)
Income			-0.006
			(0.475)
Pol. orientation			-0.527
			(0.539)
Trust in AI			0.453
			(0.565)
Favorable to China			0.068
			(0.901)
UK			-0.724
			(1.827)
Canada			0.409
			(1.875)
Observations	1017	1017	1017
R^2	0.017	0.199	0.207

Notes: OLS estimates. Dependent variable: change in perceived bias against China (post − pre, 0–100 scale). Baseline: Static. Robust standard errors in parentheses. Column (1): unadjusted. Column (2): pre-treatment bias control. Column (3): demographic controls and country fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.2 Secondary Outcomes

Table 4 reports effects on all four outcomes using the preferred specification (OLS with pre-treatment control). Both Chatbot and Comment significantly increase perceived one-sidedness relative to Static (column 2). Neither treatment significantly affects trustworthiness or sharing intentions.

Table 4: Treatment Effects on All Outcomes

	Change Score (Post − Pre)			
	(1) Trust	(2) One-sided	(3) Bias	(4) Share
Chatbot	-1.128	5.431***	8.515***	0.106
	(1.448)	(1.808)	(1.905)	(1.222)
Comment	-1.322	5.331***	5.290***	0.314
	(1.547)	(1.884)	(1.996)	(1.257)
Observations	1017	1017	1017	1017
R^2	0.174	0.293	0.199	0.136

Notes: OLS estimates. Each column reports a separate regression of the change score on treatment indicators and the pre-treatment level of the outcome. Baseline: Static. Robust standard errors in parentheses. Primary outcome: perceived bias (column 3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 2 visualizes the treatment effects across all outcomes.

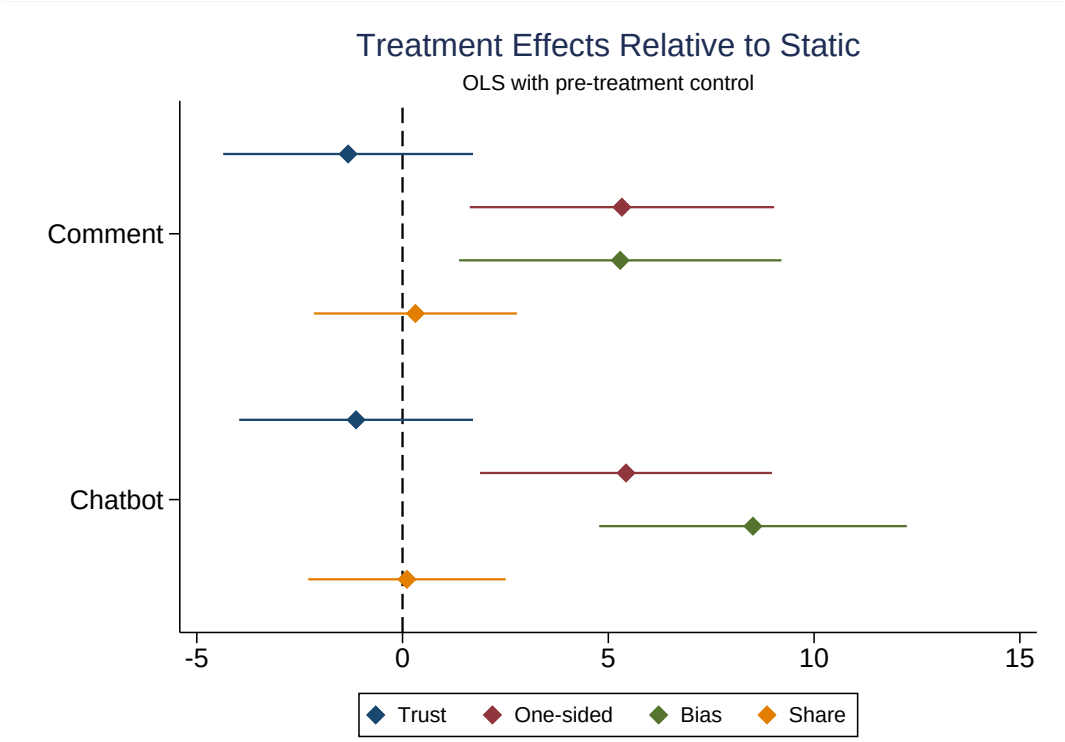

Figure 2: Treatment Effects Relative to Static

3.3 Robustness: Permutation Tests

We verify the main results using Fisher-Pitman permutation tests (200,000 Monte Carlo simulations). Table 5 reports two-tailed p-values.

Table 5: Pairwise Permutation Tests (p-values, Two-Tailed)

Outcome	Chatbot vs. Static	Comment vs. Static	Chatbot vs. Comment
Bias (primary)	**<0.001**	**0.013**	0.098
Trust	0.445	0.443	0.945
One-sidedness	**0.007**	**0.008**	0.917
Sharing	0.923	0.809	0.691

Notes: Each cell reports a two-tailed p-value from a Fisher-Pitman permutation test (200,000 Monte Carlo simulations) comparing the mean change score between two treatment groups. Bold indicates $p < 0.05$.

The primary result—Chatbot significantly increases perceived bias relative to Static—is confirmed nonparametrically ($p < 0.001$). The Comment vs. Static contrast is also significant ($p = 0.013$). Both Chatbot and Comment significantly increase perceived one-sidedness ($p < 0.01$).

3.4 Multiple Hypothesis Testing Corrections

We address multiplicity along two dimensions. First, we apply Bonferroni-Holm across the three pairwise treatment comparisons on the primary outcome (Panel A of Table 6). Second, we apply the Romano-Wolf stepdown procedure (Romano & Wolf, 2005) across the four-outcome family for each treatment contrast (Panel B), using 5,000 bootstrap replications.

Table 6: Multiple Hypothesis Testing Corrections

	Unadjusted	Bonferroni-Holm
Panel A: Three pairwise comparisons on perceived bias		
Chatbot vs. Static	<0.001	<0.001
Comment vs. Static	0.008	0.016
Chatbot vs. Comment	0.056	0.056
	Unadjusted	Romano-Wolf
Panel B: Four outcomes (Romano-Wolf stepdown)		
Chatbot vs. Static		
Bias	<0.001	0.0002
One-sidedness	0.002	0.007
Trust	0.420	0.637
Sharing	0.932	0.931
Comment vs. Static		
Bias	0.007	0.018
One-sidedness	0.005	0.017
Trust	0.412	0.642
Sharing	0.790	0.796

Notes: Panel A applies the Holm (1979) step-down procedure across three pairwise comparisons on the primary outcome (perceived bias). Panel B applies the Romano-Wolf (2005) stepdown procedure across four outcomes for each treatment contrast, with 5,000 bootstrap replications and robust standard errors. All regressions control for the pre-treatment level of the outcome.

Both the Chatbot and Comment effects on perceived bias survive all corrections. In Panel A, Chatbot vs. Static and Comment vs. Static are significant after Bonferroni-Holm ($p < 0.001$ and $p = 0.016$). In Panel B, the Romano-Wolf adjusted p-values confirm that the bias result is not an artifact of testing multiple outcomes: the adjusted p-values are 0.0002 (Chatbot) and 0.018 (Comment). The one-sidedness result also survives Romano-Wolf correction for both contrasts ($p \leq 0.017$).

Figure 3 shows the full distribution of change scores by treatment.

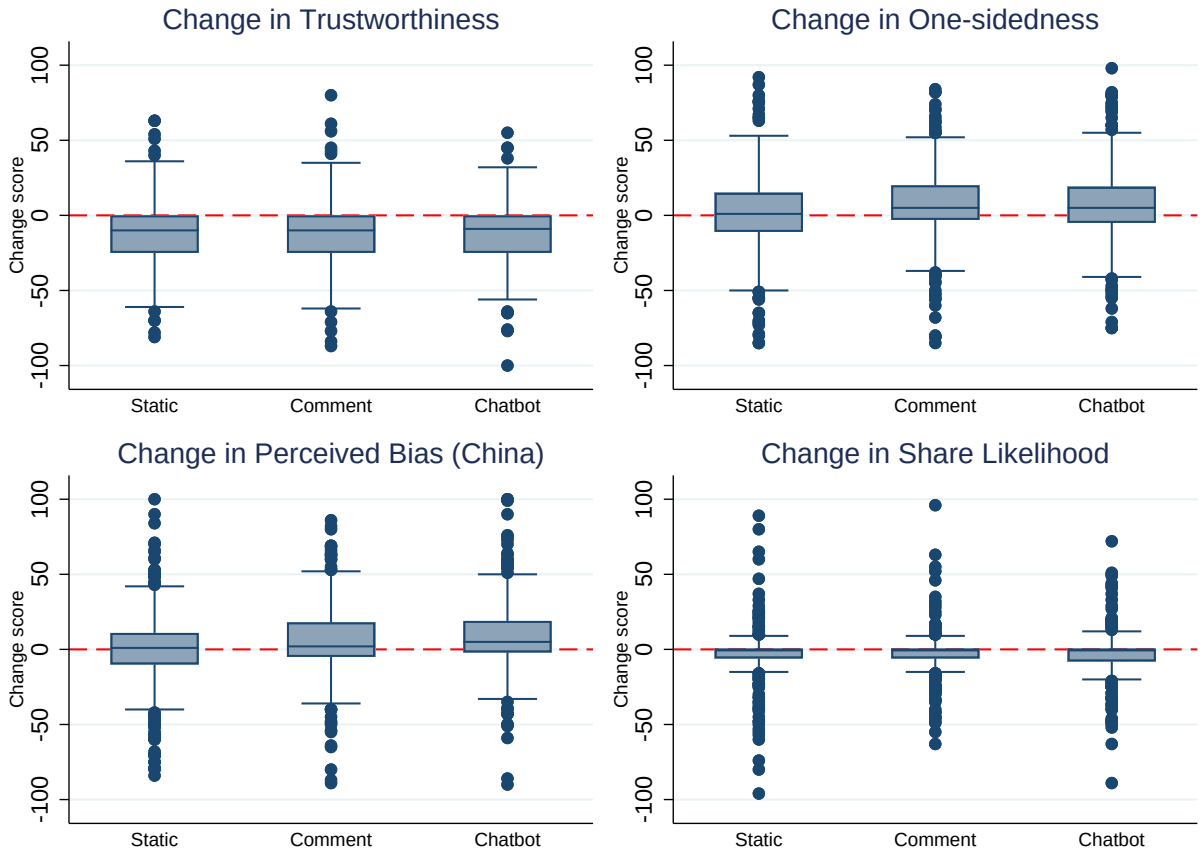

Figure 3: Distribution of Change Scores by Treatment

4 Country Heterogeneity

Table 7 reports treatment effects on the primary outcome interacted with country indicators (US is the baseline). None of the interaction terms are statistically significant, indicating that the treatment effects are consistent across the US, UK, and Canada. The point estimate for the Comment × Canada interaction is positive (+6.98) but not significant at conventional levels.

5 Exploratory Analysis: Information Processing

We examine whether three post-treatment information-processing measures predict perception changes differently across treatment arms. The three measures, each on a 1–7 Likert scale, are: (i) *fairness evaluation* ("helped me evaluate whether the article was fair"), (ii) *information novelty* ("the information provided is new to me"), and (iii) *verification motive*

Table 7: Treatment Effects on Perceived Bias by Country

	Change in Perceived Bias
Chatbot	8.744***
	(3.350)
Comment	1.469
	(3.603)
UK	-2.241
	(3.888)
Canada	-0.531
	(3.818)
Chatbot × UK	0.122
	(4.692)
Comment × UK	4.131
	(5.017)
Chatbot × Canada	-0.954
	(4.730)
Comment × Canada	6.982
	(4.885)
Observations	1017
R^2	0.204

Notes: OLS estimates. Dependent variable: change in perceived bias against China (0–100). Controls for pre-treatment bias. Baseline: Static in the US. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

("made me want to verify the article further"). Because these variables are measured after treatment, the results should be interpreted as exploratory within-treatment associations, not as causal mediation.

Table 8 reports OLS regressions of the change in perceived bias on the three information-processing measures, estimated separately within each treatment arm. The verification motive is positively associated with increased perceived bias in all three arms, but the association is strongest and most precisely estimated in the Comment and Chatbot arms. Information novelty shows no significant association in any arm.

Table 8: Information Processing and Bias Perception Change by Treatment

	Change in Perceived Bias (0–100)		
	(1) Static	(2) Comment	(3) Chatbot
Fairness evaluation	2.509*	0.796	1.590
	(1.300)	(1.153)	(1.103)
Information novelty	-0.476	-0.686	-0.731
	(1.146)	(0.837)	(0.851)
Verification motive	1.252	3.313***	2.189***
	(1.291)	(1.053)	(0.743)
Observations	294	331	392
R^2	0.215	0.254	0.205

Notes: OLS estimates within each treatment arm. Dependent variable: change in perceived bias against China (0–100). All regressions control for pre-treatment bias. All information-processing measures are on a 1–7 Likert scale. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4 displays the means of the three information-processing measures by treatment group.

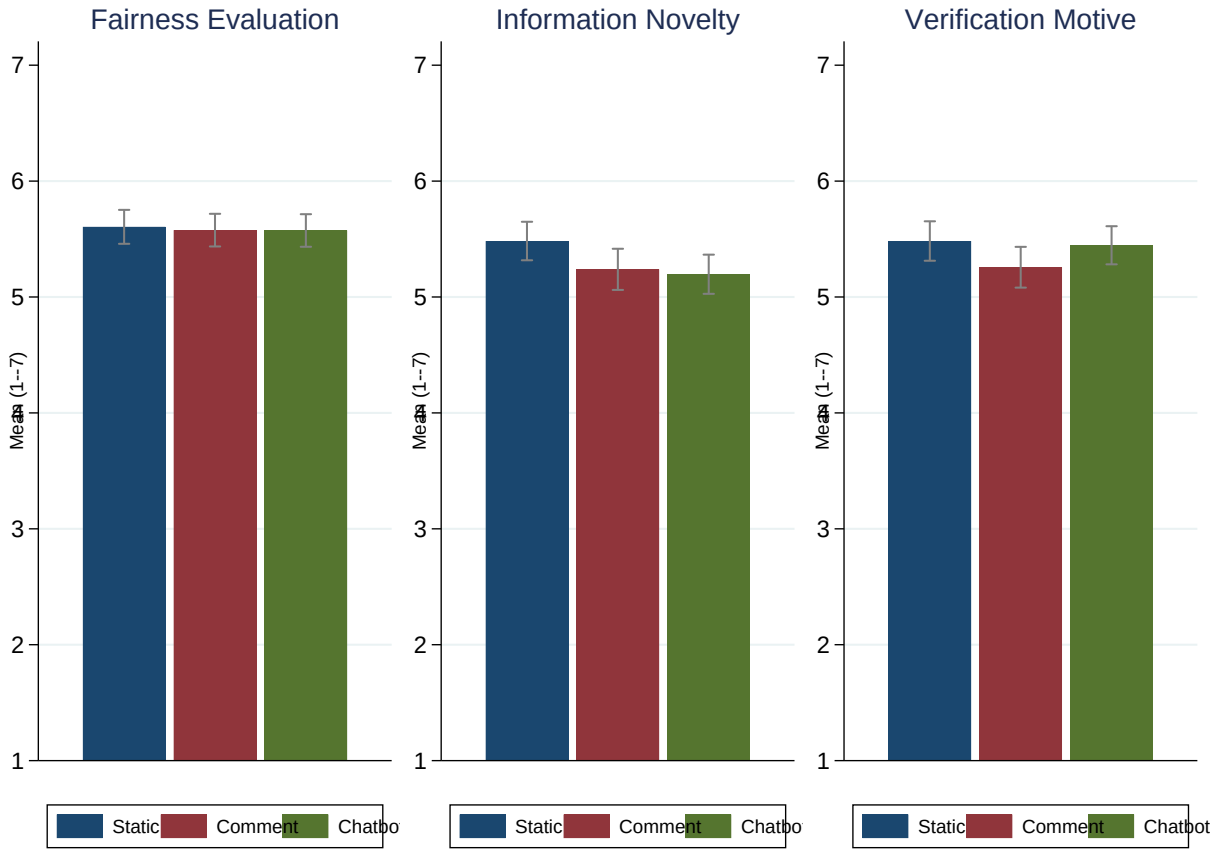

Figure 4: Information Processing Measures by Treatment (with 95% CIs)

6 Summary

Table 9: Summary of Primary and Secondary Results

Outcome	Contrast	OLS $\hat{\beta}$ (p)	With controls $\hat{\beta}$ (p)	Permutation p
Primary				
Bias	Chatbot vs. Static	8.52 (<0.001)	7.79 (<0.001)	<0.001
Bias	Comment vs. Static	5.27 (0.008)	5.02 (0.012)	0.013
Secondary				
One-sided	Chatbot vs. Static	5.43 (0.003)	—	0.007
One-sided	Comment vs. Static	5.33 (0.005)	—	0.008
Trust	Chatbot vs. Static	−1.13 (0.436)	—	0.445
Share	Chatbot vs. Static	0.11 (0.931)	—	0.923

Notes: OLS estimates from regressions controlling for pre-treatment outcome level. "With controls" adds demographics and country fixed effects. Static is the omitted baseline. Primary contrasts survive Bonferroni-Holm (pairwise) and Romano-Wolf (outcome family) corrections.

The central finding is that both AI-based interventions—the interactive chatbot and the tailored comment—significantly increase perceived bias against China relative to a static message. With the expanded three-country sample ($N = 1{,}017$), the Comment effect, which was insignificant in the two-country sample, now reaches significance ($p = 0.008$; Westfall-Young adjusted $p = 0.015$). Both interventions also increase perceived one-sidedness, but neither differentially affects trustworthiness or sharing intentions. Treatment effects are consistent across the US, UK, and Canada.

We emphasize two important limitations. First, the experiment identifies shifts in *perceived* bias, not whether the induced perceptions are accurate or welfare-improving. Second, the design cannot separate the effect of interactivity from the effect of message content: the chatbot was designed to highlight potential anti-China bias in the article, so the estimated effect reflects exposure to that framing.